

TrustLayer — AI Governance Audit Report

Smart India Hackathon 2026 | Continuous Responsible AI Assurance & Policy Enforcement

1. Model Card & Metadata

Model Name:	Loan Approval Classifier	Domain:	N/A
Version / ID:	1.0 (model-loan-01)	Target Variable:	approved
Owner / Team:	Risk Assessment Team	Sensitive Fields:	gender, age
Features:			

2. Policy Engine Verdict

POLICY VERDICT: BLOCK

- Policy violation (BLOCK): 'fairness.demographic_parity_gap' value 0.328 violated condition '<= 0.15'
- Policy violation (BLOCK): 'fairness.disparate_impact_ratio' value 0.561 violated condition '>= 0.8'

3. Human Review & Promotions

STATUS: PENDING REVIEW

This model is currently undergoing continuous validation. Promotion to the production cluster remains BLOCKED until a designated human auditor submits a signed override/approval.

4. Core Audit Metrics Summary

A. Predictive Performance

Accuracy	0.874	Precision	0.904
Recall (TPR)	0.880	F1-Score	0.892
ROC-AUC	0.939	Confusion Matrix	TN: 1052 FP: 166 FN: 213 TP: 1569

B. Demographic Group Fairness

Sensitive Attribute	gender	Demographic Parity Gap	0.328
Disparate Impact Ratio	0.561	Equal Opportunity Gap (TPR)	0.094
Selection Rates	Female: 42.0%, Male: 74.8%	Fairness Status	FAIL

5. Explainability & Feature Importances (SHAP)

Rank	Feature Name	SHAP Importance (Relative)
1	credit_score	0.38
2	income	0.28
3	debt_ratio	0.18
4	employment_years	0.11
5	age	0.05

6. Compliance & Standards Alignment Framework

Standard Framework	Mapped TrustLayer Evidence Support
NIST AI Risk Management Framework (AI RMF 1.0)	Audit tracks 'Fair' and 'Explainable' pillars. Disparity evaluations (selection rates, TPR gaps) map directly to RMF Measure 1.2 & 2.1.
ISO/IEC 42001:2023 (AI Management System)	Continuous verification logs provide documented evidence of system control metrics (Annex A.3 - Impact Assessment).
Regulatory Directives (EU AI Act / India DPDP Act)	Preserves immutable model card artifacts and human sign-off history to demonstrate automated decision governance accountability.